

CIRCLE THEOREMS

Higher Tier

Angles and problem solving

Name

Date

13 questions

52 marks

2017-2025

Instructions

- Answer each question in the space provided.
- Show all your working and follow the rounding instructions.
- Diagrams are not accurately drawn unless stated otherwise.
- Use a calculator except for pages marked NON-CALCULATOR.

Using this booklet

Questions are arranged by exam series. Each page gives its original paper reference and question number. The question numbers within the extracted artwork are the original exam numbers.

This is an independently compiled practice resource, not an official Pearson examination paper. The source collection covers 2017-2025. Questions are included only in the booklet that matches their main purpose.

Question index

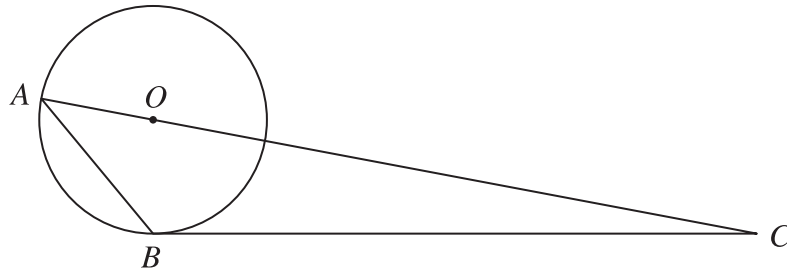
Higher Tier | 13 questions | 52 marks

Practice	Exam series	Paper	Original question	Marks	Page
01	June 2018	1MA1/1H	Q11	5	3
02	November 2018	1MA1/1H	Q12	3	4
03	June 2019	1MA1/2H	Q18	5	5
04	November 2019	1MA1/2H	Q17	3	6
05	November 2020	1MA1/3H	Q14	4	7
06	November 2021	1MA1/2H	Q14	4	8
07	June 2022	1MA1/2H	Q20	4	9
08	November 2022	1MA1/3H	Q16	4	10
09	June 2023	1MA1/1H	Q21	4	11
10	November 2023	1MA1/3H	Q21	4	12
11	November 2024	1MA1/1H	Q17	4	13
12	June 2025	1MA1/3H	Q20	4	14
13	November 2025	1MA1/3H	Q13	4	15

The original exam numbering is retained in each question.

Paper 1 questions are marked NON-CALCULATOR on their pages.

11



A and B are points on a circle, centre O .

BC is a tangent to the circle.

AOC is a straight line.

Angle $ABO = x^\circ$.

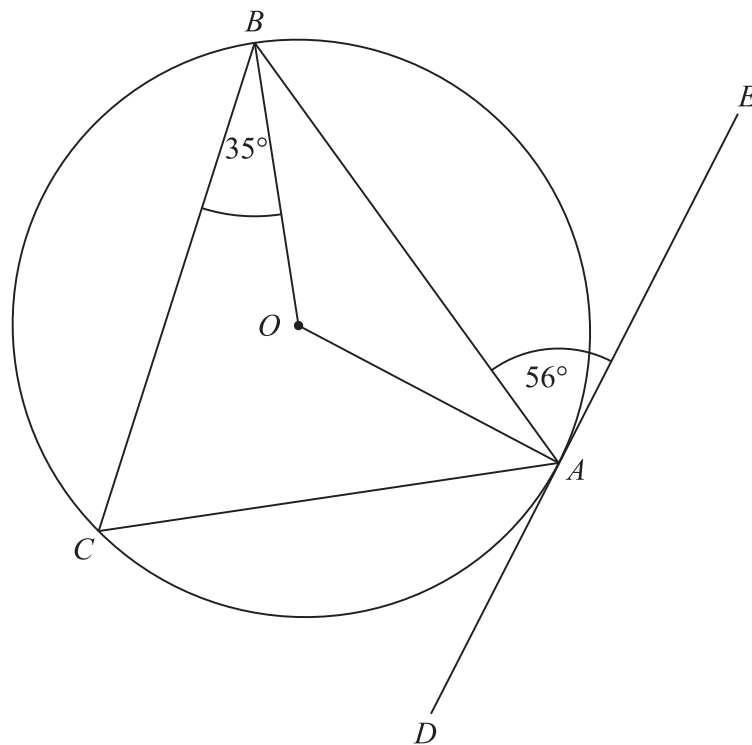
Find the size of angle ACB , in terms of x .

Give your answer in its simplest form.

Give reasons for each stage of your working.

(Total for Question 11 is 5 marks)

12



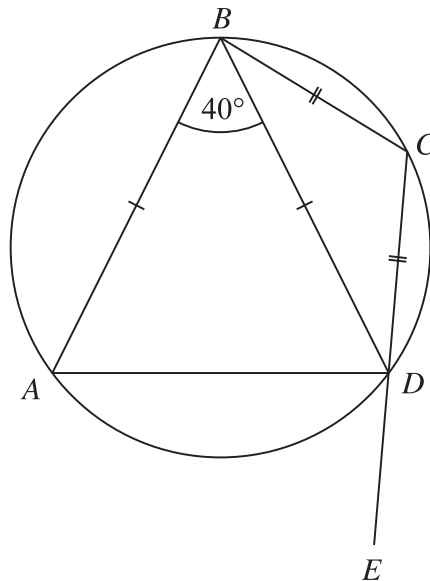
A , B and C are points on the circumference of a circle, centre O .
 DAE is the tangent to the circle at A .

Angle $BAE = 56^\circ$
Angle $CBO = 35^\circ$

Work out the size of angle CAO .
You must show all your working.

(Total for Question 12 is 3 marks)

- 18 The points A , B , C and D lie on a circle.
 CDE is a straight line.



$$BA = BD$$

$$CB = CD$$

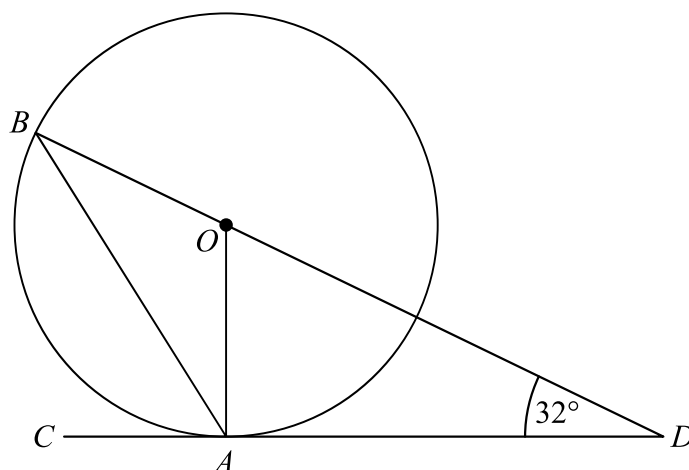
$$\text{Angle } ABD = 40^\circ$$

Work out the size of angle ADE .

You must give a reason for each stage of your working.

(Total for Question 18 is 5 marks)

17



A and B are points on a circle with centre O .

CAD is the tangent to the circle at A .

BOD is a straight line.

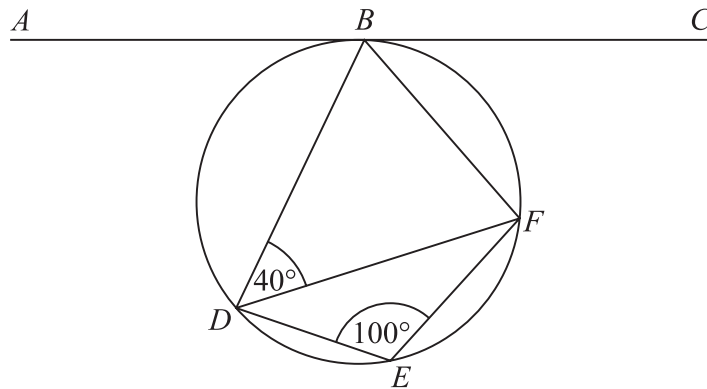
Angle $ODA = 32^\circ$.

Work out the size of angle CAB .

You must show all your working.

(Total for Question 17 is 3 marks)

14

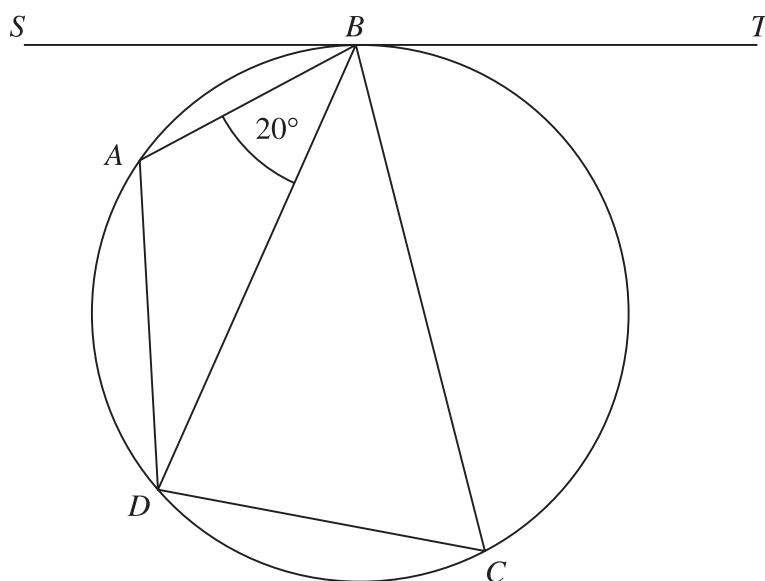


Points B , D , E and F lie on a circle.
 ABC is the tangent to the circle at B .

Find the size of angle ABD .
You must give a reason for each stage of your working.

(Total for Question 14 is 4 marks)

14



A , B , C and D are four points on a circle.

SBT is a tangent to the circle.

Angle $ABD = 20^\circ$

the size of angle BAD :the size of angle $BCD = 3 : 1$

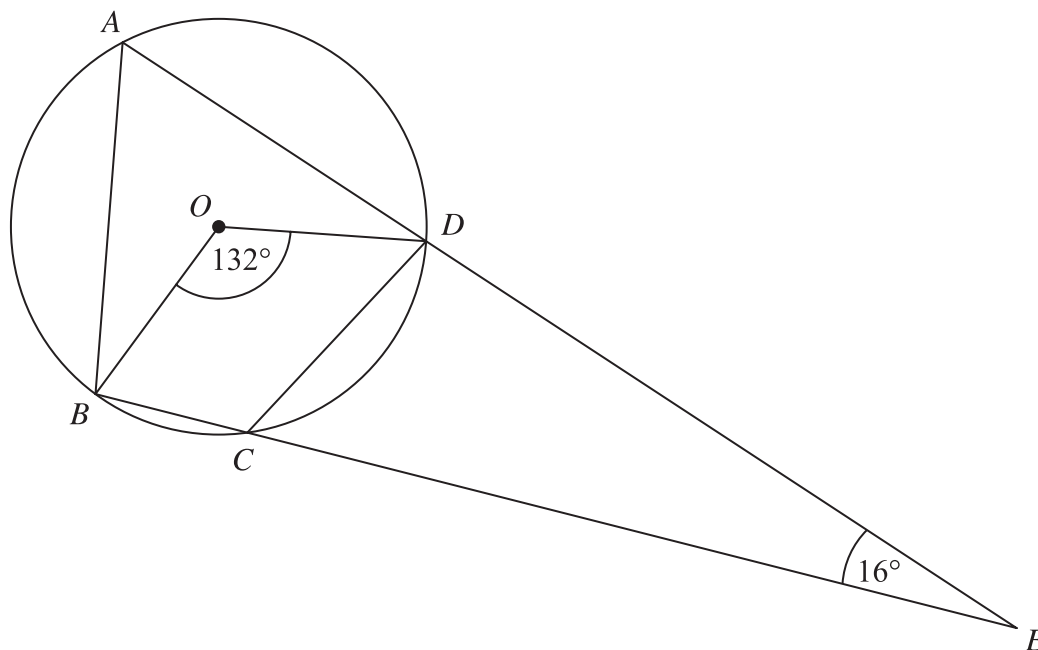
Find the size of angle SBA .

Give a reason for each stage of your working.

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(Total for Question 14 is 4 marks)

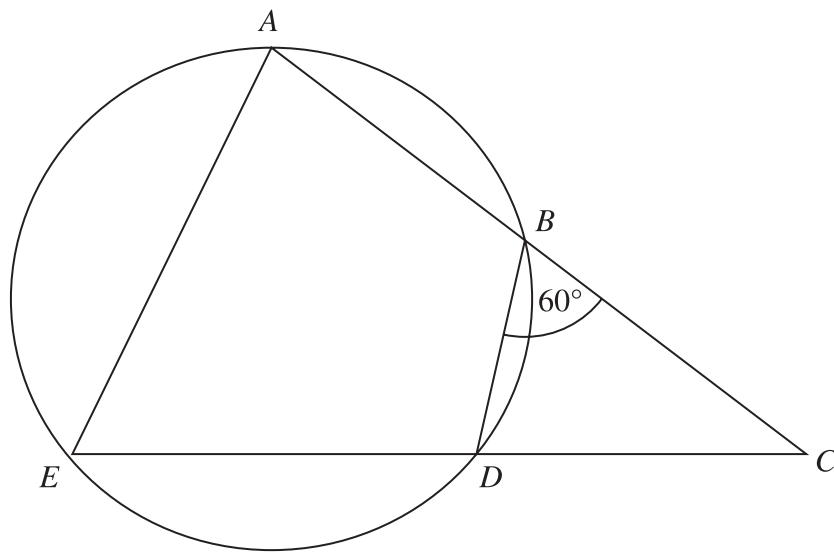
- 20 A , B , C and D are points on the circumference of a circle, centre O .
 ADE and BCE are straight lines.



Work out the size of angle CDE .
Give a reason for each stage of your working.

(Total for Question 20 is 4 marks)

16



$ABDE$ is a cyclic quadrilateral.
 ABC and EDC are straight lines.
Angle $DBC = 60^\circ$

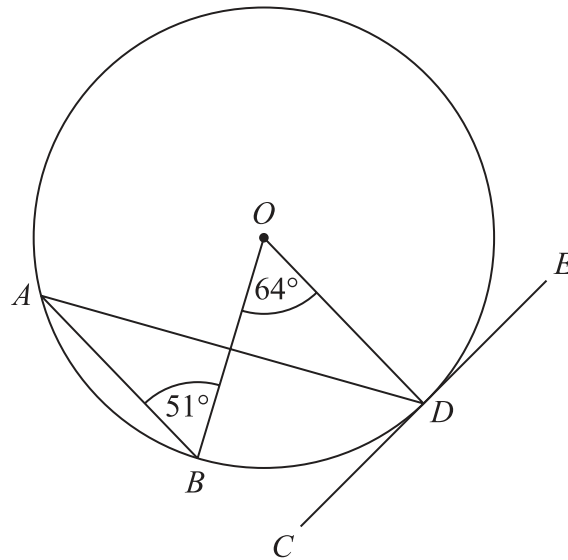
Given that

$$\text{size of angle } EAB : \text{size of angle } BCD = 2 : 1$$

work out the size of angle BCD .
You must show all your working.

(Total for Question 16 is 4 marks)

- 21 A , B and D are points on a circle with centre O .
 CDE is the tangent to the circle at D .

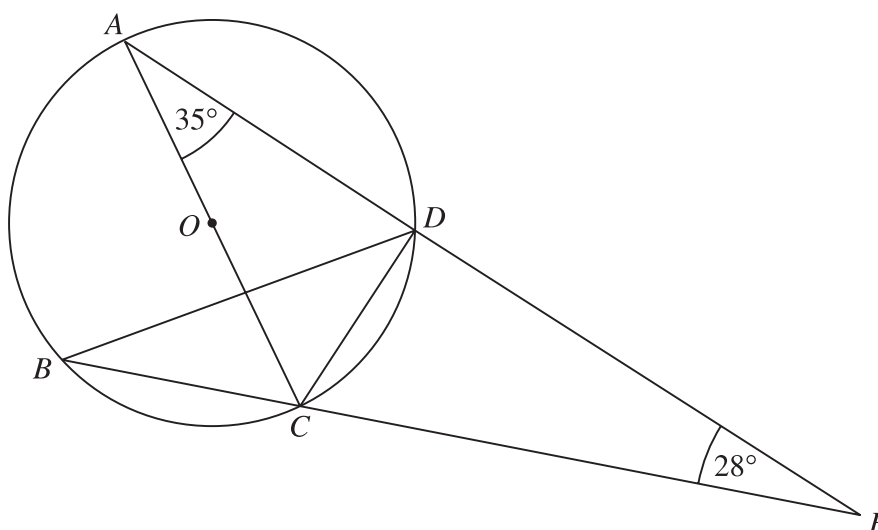


Work out the size of angle ADC .
Write down any circle theorems you use.

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(Total for Question 21 is 4 marks)

21



A , B , C and D are points on the circumference of a circle, centre O .
 AC is a diameter of the circle.

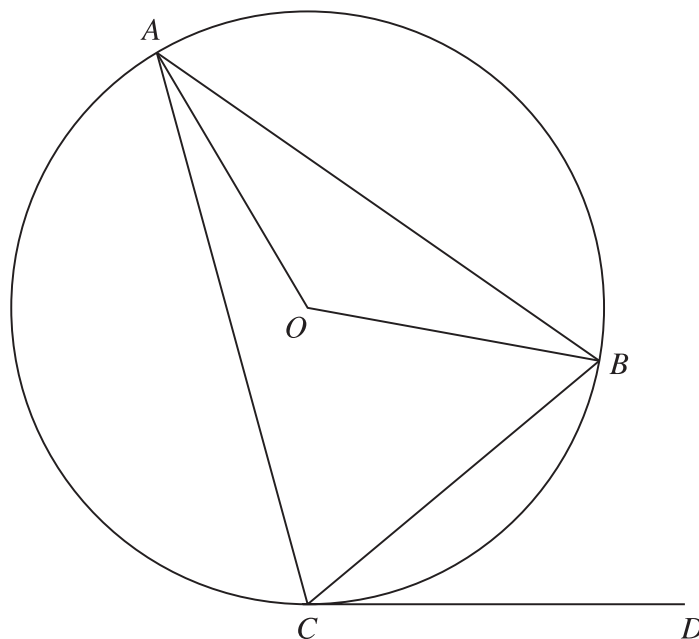
ADE and BCE are straight lines.

Work out the size of angle BDC .

Write down any circle theorems that you use.

(Total for Question 21 is 4 marks)

17 A , B and C are points on a circle, centre O .



CD is a tangent to the circle.

Angle $BCD = 40^\circ$

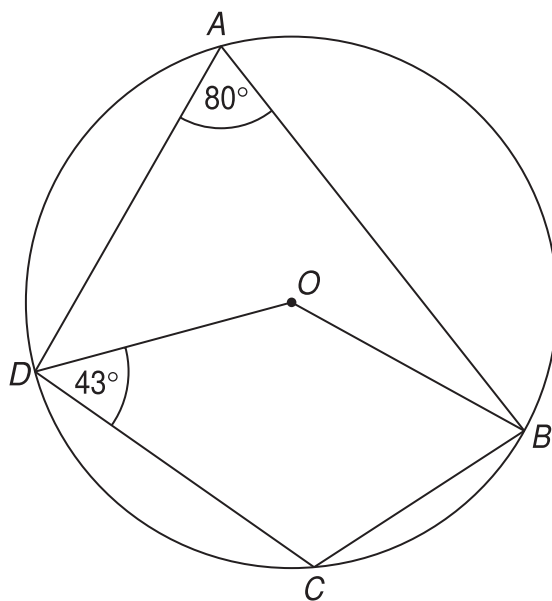
Angle $OAB = 3 \times \text{angle } OAC$

Work out the size of angle ACD .

Write down any circle theorems that you use.

(Total for Question 17 is 4 marks)

20 A , B , C and D are points on a circle with centre O .

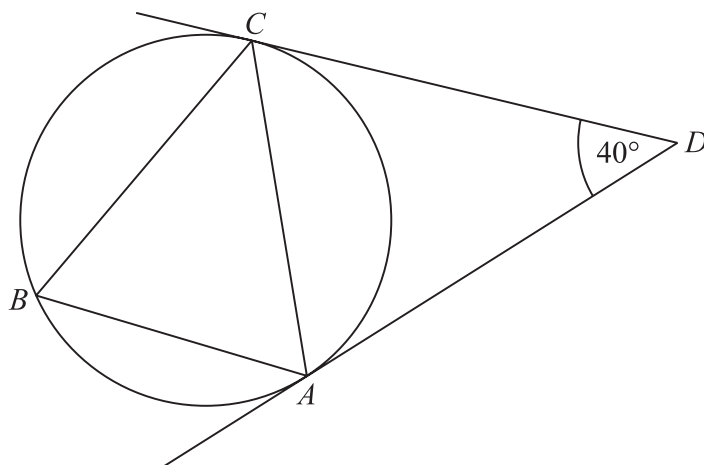


Find the size of angle OBC .

Write down any circle theorems that you use.

(Total for Question 20 is 4 marks)

13 A , B and C are three points on a circle.



CD and AD are tangents to the circle.

Find the size of angle ABC .

You must give a reason for each stage of your working.

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(Total for Question 13 is 4 marks)